

Mark Verma

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EXPERIENCE

314e Corp

Software Development Engineer

Associate Software Development Engineer

Software Development Engineer Intern

Bengaluru, India

Jan 2025 – Present

Dec 2024 – Jan 2025

Jul 2024 – Dec 2024

- Built and owned the entity-extraction ML stack behind a clinical document-processing product, taking a fine-tuned 27B vision-language model to 0.96 test accuracy across 26 clinical entity types in production
- Wrote a 7,600-line model-agnostic VLM fine-tuning package on SkyPilot, Temporal and ClearML that trains Gemma 3, Qwen3.5-9B and Qwen3.6-27B through one model-family dispatch layer, rewritten across three generations as the model family changed
- Built the human-in-the-loop retraining loop from reviewer feedback through stratified dataset construction to automated deployment, gated by a per-entity F1 regression check that blocks any model regressing beyond a tolerance scaled to its prior score
- Built a weekly accuracy-drift monitor separating genuine model regression from reviewer labelling-convention change, two causes indistinguishable in F1 that demand opposite responses, using an LLM agent over a purpose-built read-only MCP server that reads page OCR to check the document itself
- Wrote the schema-driven entity-extraction layer and a per-field confidence tree mapping token logprobs to sequence likelihood, spanning 4 LLM providers and 3 inference engines behind one validated output contract
- Trained a calibrated XGBoost confidence model over those logprobs, OCR grounding and text embeddings on 99,427 reviewed extractions, halving the false-approval rate against the LLM's own confidence at matched coverage (1.4% vs 2.7%) and calibrating expected error from 0.139 to 0.004, retiring 83% of manual entity review
- Migrated entity-extraction serving from Transformers to vLLM to SGLang on RunPod serverless, reaching 64 concurrent jobs per worker with continuous batching, and removed cold-start cost by baking weights into the image and pre-warming triton kernel JIT and vision-encoder init
- Cut model training time from 90 hours to 25 hours by replacing an iterable dataset with map-style lazy loading and sizing batches from available VRAM, after tracing an 80 GB host-RAM exhaustion to a cap that bounded sample count but never bytes

Banach Technologies

Machine Learning Intern

Singapore (Remote)

Apr 2024 – Jul 2024

- Built and deployed ML models against live market APIs to execute hedging strategies, cutting data-processing latency through targeted performance tuning and covering the core trading libraries with unit tests

Alemeno

Software and ML Intern

Maharashtra (Remote)

Jul 2023 – Apr 2024

- Built a preprocessing pipeline for semantic-segmentation models over large-scale GIS raster data, implementing distributed computer-vision algorithms including Douglas-Peucker and Jarvis March, and deployed Django services on AWS behind end-to-end CI/CD

Bewgle

Data Analyst Intern

Bengaluru (Remote)

May 2022 – Jul 2022

- Applied NLP to proprietary Amazon review datasets for product-trend insight, automating preprocessing in Python and Bash to cut data redundancy 90%

PROJECTS

Resume Check — TypeScript, Transformers.js, ONNX Runtime

github.com/markvrma/resume-parser

- Built a browser-only resume scorer that parses PDFs client-side and never uploads the file, choosing NLI over embedding cosine similarity after measuring cosine rank an unmet requirement above a met one (0.388 vs 0.215) where NLI separated them cleanly, and selecting WASM over WebGPU at 49 ms vs 660 ms per forward pass

rumpy — Rust

github.com/markvrma/rumpy

- Built a zero-dependency ndarray library in Rust on a flat-buffer, shape and strides layout, reimplementing broadcasting, axis reductions and cache-friendly ikj matmul, with 30 milestone tests benchmarked against NumPy

OPEN SOURCE

- Authored 8 pull requests across PyMC, MDAnalysis and sktime with 4 merged, including generalizing the PyMC Multinomial moment computation to arbitrary batch dimensions in pymc-devs/pymc#5476 and removing 91 lines of dead code; MDAnalysis work completed under Google Season of Docs
- Merged 19 of 21 pull requests to a collaborative Discord bot, adding a vote-to-skip queue and Billboard Hot 100 integration and fixing a recurring API-timeout defect, placing 2nd in an open-source competition (github.com/markvrma/discord-bot-sample)

PUBLICATION

Trained a CNN to 98.4% accuracy for medical-waste classification and deployed it as an interactive TensorFlow.js web application, published as first author in IEEE IATMSI 2022 and cited 25 times (ieeexplore.ieee.org/document/10119431)

SKILLS

Languages: Python, C++, Rust, SQL, Bash

ML and LLM: PyTorch, Hugging Face Transformers, vLLM, SGLang, XGBoost, LoRA fine-tuning, vision-language models, structured outputs, probability calibration, inference optimization, model evaluation, drift detection, SHAP, RAG, agentic tool use, Model Context Protocol, NumPy, Pandas, scikit-learn

Infrastructure: Docker, Kubernetes, Ray Serve, Temporal, SkyPilot, ClearML, MLflow, RunPod, FastAPI, gRPC, PostgreSQL, AWS, Git, Linux

EDUCATION

IIITM Gwalior — B.Tech, Computer Science, SGPA 8.67

Gwalior, India • Graduated 2024